Intelligence Retail Ecosystem

Ambient Intelligence Alameda G11

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1. Introduction

This project develops an Ambient Intelligence (AmI) ecosystem to bridge the gap between digital inventory and physical navigation. By integrating BLE infrastructure and computer vision (CV), the solution provides high-precision guidance in high-density retail environments like IKEA or supermarkets.

Significance

Shoppers often face "search friction" in massive layouts. Our system eliminates this by transforming the store into an interactive assistant. Users simply ask their phone for a product, and the system provides a real-time path followed by a visual shelf highlight. This "search-to-shelf" automation reduces cognitive load and ensures a frictionless experience for all users.

Objectives

- **Precision Localization:** Achieve <1.5 m accuracy using a hybrid engine that fuses BLE trilateration with IMU data and Kalman filtering to overcome shelf interference.
- **Seamless Navigation:** Provide real-time pathfinding via a web-based digital map.
- **Micro-Navigation:** Display a real-world photo of the target shelf with the specific product digitally highlighted upon arrival.
- **Automated Inventory:** Utilize CV modules to automatically map product positions (height/aisle) through photo analysis, maintaining a real-time "digital twin" of the store.

2. Literature Review

Paper 1: Obeidat, H., et al. (2023). "On Indoor Localization Using WiFi, BLE, UWB, and IMU Technologies."

Main Concepts: This technical study evaluates the four primary wireless technologies used for indoor positioning, with a specific focus on their performance in "cluttered" environments. It introduces the concept of data fusion, which integrates radio signals with inertial measurement units (IMUs) like accelerometers and gyroscopes.

Strengths & Weaknesses: The major strength of this work is its comparison of signal propagation physics. It highlights that while UWB offers centimeter-level precision, it requires expensive, specialized hardware for both the infrastructure and the user. Conversely, BLE is identified as the most scalable and cost-effective solution for public spaces due to its universal presence in smartphones. A noted weakness is the "noise" or multipath interference BLE faces in environments with high metal density (like warehouses), which can cause location "jumping."

Influence on Proposal: This paper is the technical foundation for our hybrid localization method. Following the authors' conclusion that "hybrid" systems provide the best balance for retail, we have adopted the use of Pedestrian Dead Reckoning (PDR) alongside BLE. This compensates for the signal fluctuations discussed in the paper, allowing us to aim for a very small margin of error essential for finding small items on a shelf.

Paper 2: Ayoola, V.B., et al. (2024). "IoT-driven Smart Warehouses with Computer Vision for Enhancing Inventory Accuracy and Reducing Discrepancies in Automated Systems."

Main Concepts: This review paper examines the use of IoT infrastructure and computer vision within automated warehouse environments. It explores how interconnected devices including RFID tags, environmental sensors, and IoT gateways can be paired with vision-based systems to automate product recognition, item counting, and anomaly detection, delivering real-time inventory visibility across the supply chain.

Strengths & Weaknesses: The authors convincingly demonstrate that combining these technologies directly addresses chronic warehouse problems such as miscounts, misplaced items, and human error. However, the paper is primarily a literature review rather than a study, meaning it does not provide original performance benchmarks or experimental validation of the proposed integrations. It also openly acknowledges unresolved challenges around system integration complexity, data management at scale, and the cost of deployment for smaller operations.

Influence on Proposal: This paper supports our inventory management layer. Its argument that computer vision and IoT sensing are most powerful when tightly integrated rather than deployed in isolation directly reinforces our architectural decision to couple real-time shelf-scanning with an IoT-connected backend. The scalability and data management challenges it flags also serve as a design constraint for our system, pushing us toward a lightweight, modular implementation that can be realistically deployed.

3. Problem

The specific problem being solved is the "information-location gap" in large-scale indoor environments. While digital inventories know *what* is in stock, they often fail to communicate *exactly where* the item is to a moving human in a way that is intuitive. In spaces like IKEA or large supermarkets, shoppers face "search fatigue" due to complex layouts and similar-looking packaging, while staff lose significant time performing manual stock-location audits.

Stakeholders:

- **Shoppers/Visitors:** Individuals looking for specific items who want to minimize time spent wandering.
- **Retail/Warehouse Staff:** Workers responsible for picking orders and maintaining accurate stock location data.
- **Store Managers:** Business owners seeking to increase sales conversion and reduce labor costs associated with inventory management.

Value Added:

- **Efficiency:** Reduces "time-to-item" for shoppers, leading to higher satisfaction and faster throughput.
- **Accuracy:** Eliminates the discrepancy between the digital inventory list and the physical shelf location.
- **Accessibility:** Provides a hands-free, voice-guided experience for users who are physically burdened or have visual impairments.

Requirements

R1: Localization & Tracking

- **R1.1: Precision:** Maintain <1.5m error to ensure accurate user positioning.
- **R1.2: Stability:** Fuse BLE + IMU data to prevent marker "jumping".
- **R1.3: Latency:** Recalculate navigation paths within 1 second of user deviation.

R2: User Interaction & Navigation

- **R2.1: Querying:** Support both natural language voice and text-based product searches.
- **R2.2: Optimization:** Generate the most efficient route for multi-product shopping lists.
- **R2.3: Wayfinding:** Display a real-time directional path on a digital floor plan.
- **R2.4: Micro-Navigation:** Show a highlighted shelf photo upon arrival at the target.

R3: Inventory & Management

- **R3.1: Staff Tools:** Provide an interface for shelf photography and automated coordinate registration.

R4: Platform & Interface

- **R4.1: Access:** Ensure compatibility with standard mobile web browsers (PWA).

3.2. Assumptions

- **Hardware:** User smartphones support Bluetooth 4.0+ and have a functional camera.
- **Infrastructure:** 3+ ESP32 gateways are elevated to minimize signal interference from obstacles.
- **Environment:** Ambient lighting of 300 lux is sufficient for computer vision accuracy.
- **Data:** A product database is pre-populated for real-time matching.

4. Prototype

4.1. Technical Overview

The prototype utilizes a Distributed Ambient Intelligence (AmI) architecture to bridge the gap between digital inventory and physical location. The system is divided into three functional layers to ensure high-precision tracking and low-latency user feedback:

- **ESP32 Gateway Network:** Three fixed listeners that scan the environment for the smartphone's BLE signature. They forward raw RSSI (signal strength) data to the backend.
- **Web Mobile Client:** A React-based Progressive Web App (PWA) that broadcasts a unique BLE session ID while simultaneously streaming IMU data (gyroscope and step counts) via WebSockets.
- **Central Backend:** A FastAPI Python server that performs sensor fusion, merging absolute BLE coordinates with relative movement data.
- **Computer Vision Module:** An AI-powered engine that analyzes shelf images to provide "micro-navigation" visual highlights.

4.2. Logical Design

1. **Hybrid Localization:** Computes user position by fusing BLE signal strength with IMU sensor data (steps/rotation) for sub-meter stability.
2. **Macro-to-Micro Navigation:** Transitions from a 2D floor map for travel to a highlighted shelf photo upon arrival for precise item retrieval.
3. **Hands-Free Search:** Enables product lookup via natural language voice commands, eliminating manual typing during navigation.
4. **Rapid Photo Inventory:** Allows staff to update the "digital twin" by simply photographing shelves to auto-register product coordinates.

4.3. Technology Selection

Component	Technology	Purpose
Frontend	React.js (PWA)	User navigation and search interface
Backend	FastAPI (Python)	RSSI processing, positioning, REST API
Database	SQLite	Product inventory and coordinates
BLE Hardware	ESP32 (x3)	Fixed BLE anchor beacons
Communication	WebSockets / MQTT	Real-time position updates

5 Bill-of-Materials (BOM)

5.1 Hardware

- **3x ESP32 Microcontrollers:** Fixed gateways used for high-frequency signal capture and MQTT data relay.

- **User Smartphone:** The central sensor hub and UI for voice/visual navigation.
- **Central Server (laptop):** Processes heavy computational tasks, position fusion, and YOLO analysis.

5.2 Software

- **React.js:** Frontend framework for the mobile PWA.
- **FastAPI:** High-performance backend for real-time pathfinding logic.
- **MQTT & WebSockets:** Protocols for low-latency hardware and sensor data streaming.
- **Web Speech API:** Native utility for hands-free, voice-activated product search.
- **YOLO:** Uses a Python-based AI engine to automatically identify products and generate visual highlights for micro-navigation
- **SQLite:** product database

6. Plan

6.1. Tasks & Responsibilities

- **T1: Infrastructure: (Gonçalo Oliveira)** Deploy and configure the 3x ESP32 gateways, set up the MQTT broker, and establish the raw BLE data pipeline to the backend server.
- **T2: Navigation & Hybrid Logic: (David Correia)** Develop the FastAPI backend, implement the BLE trilateration algorithm, integrate IMU/PDR sensor fusion with Kalman filtering, and build the SQLite spatial database for product coordinates.
- **T3: Interface & AI: (Pedro Duarte)** Build the React.js PWA, integrate the Web Speech API for voice search, implement the 2D floor map navigation view, and train/integrate the YOLOv8 model for shelf product detection and visual highlighting.

6.2. Milestones

M1: Basic (Connectivity)

- **Deliverables:** ESP32-to-server pipeline and basic 2D map.
- **Functionality:** Users track their approximate location via raw BLE signals and navigate to hardcoded product positions.

M2: Intermediate (Hybrid & Manual Visuals)

- **Deliverables:** Hybrid tracking (BLE + Gyroscope/PDR) and proximity-triggered "Shelf View."
- **Functionality:** Smooth localization using phone motion sensors. Upon arrival, the app displays a stock image of the shelf for manual product identification.

M3: Advanced (AI & Voice)

- **Deliverables: YOLO** recognition engine and Web Speech API integration.
- **Functionality:** Hands-free voice search. The system uses AI to automatically identify and digitally highlight products within the shelf image.

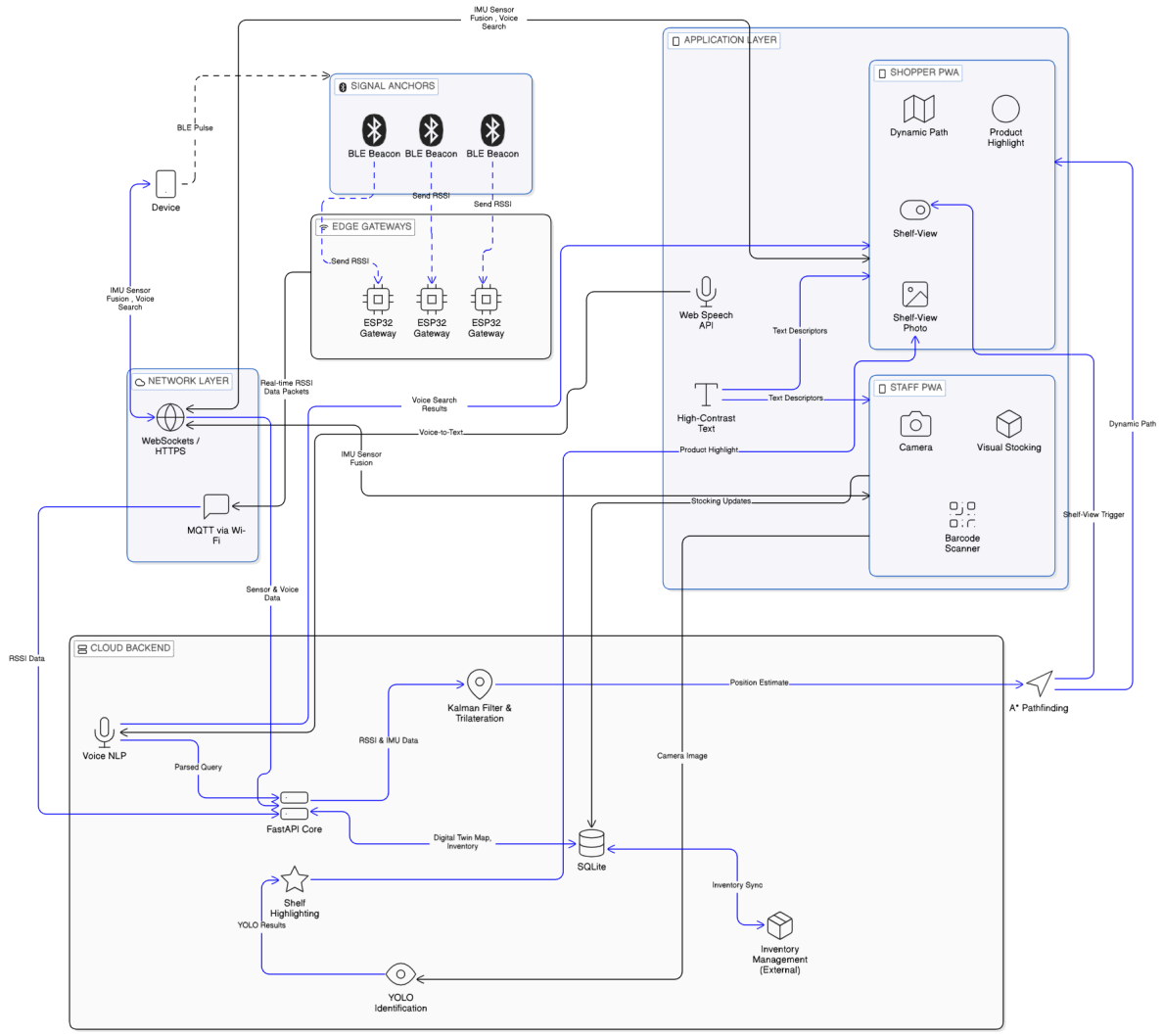
7. Bibliography

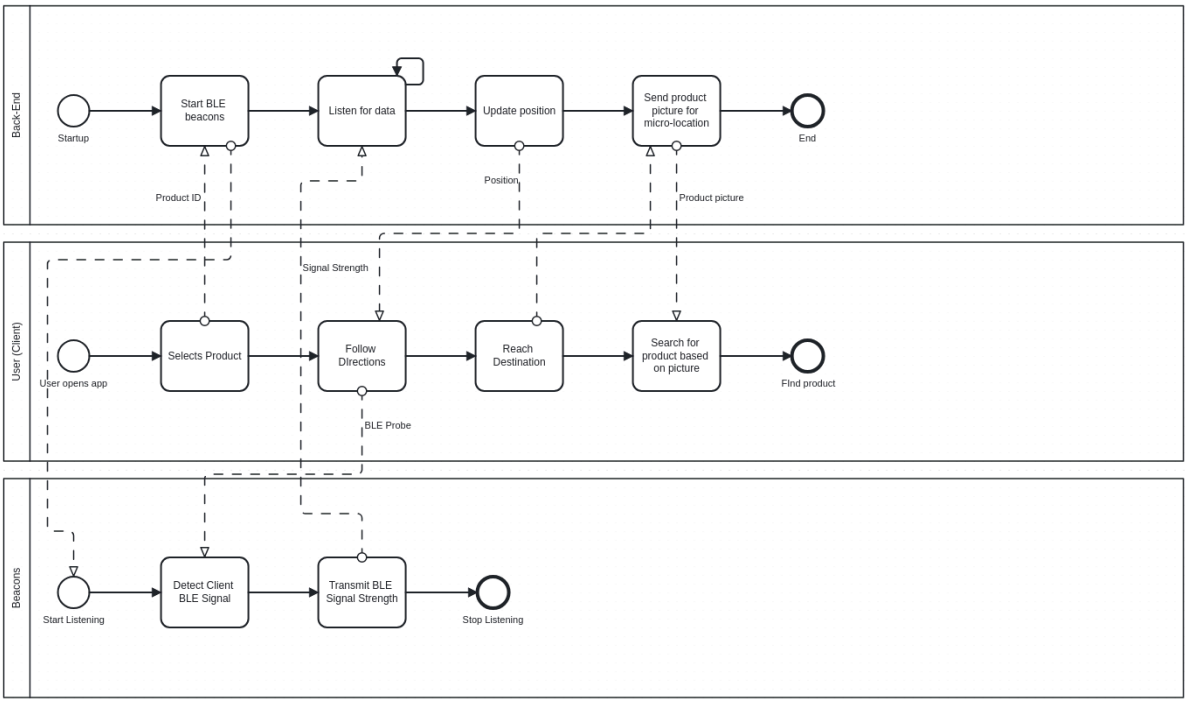
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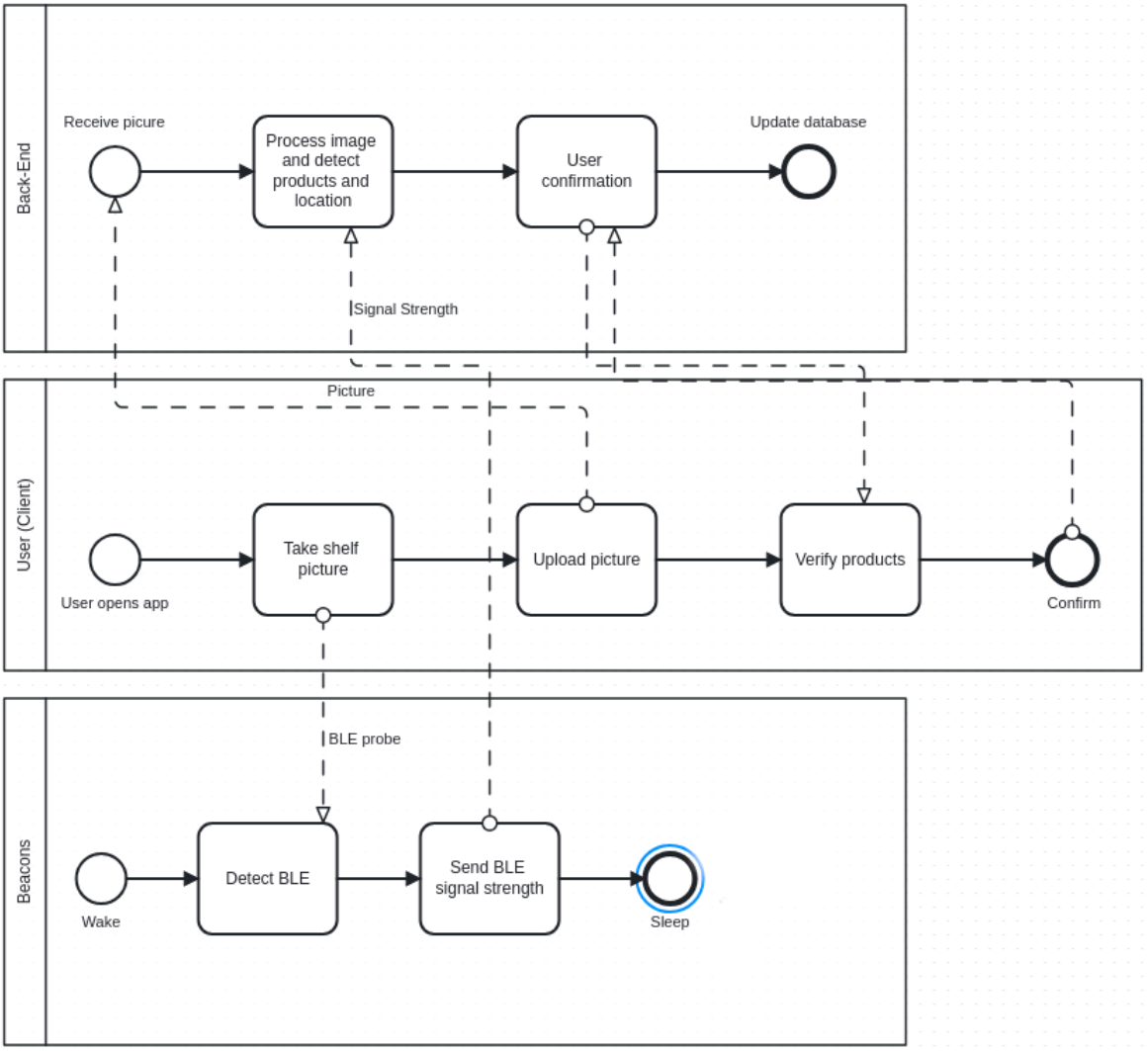
8. Declaration of Use of AI

All decisions and technical content in this work are our own. AI was used to assist in structuring this proposal and refining technical requirements.

9. Diagrams




Back-End
Startup
Start BLE beacons
Listen for data
Update position
Send product picture for micro-location
End
Product ID
Position
Product picture
Signal Strength
User (Client)
User opens app
Selects Product
Follow DIrections
Reach Destination
Search for product based on picture
Find product
BLE Probe
Beacons
Start Listening
Detect Client BLE Signal
Transmit BLE Signal Strength
Stop Listening


Back-End
Receive picure
Process image and detect products and location
User confirmation
Update database
Signal Strength
Picture
User (Client)
User opens app
Take shelf picture
Upload picture
Verify products
Confirm
BLE probe
Beacons
Wake
Detect BLE
Send BLE signal strength
Sleep